



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -3

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Control Structures

- In programming, control structures decide the *flow of execution*—that is, which statements run, when they run, and how many times they run.

There are three types of control structures:

1. Sequence
2. Selection also called as decision making/Conditional statements
3. Iteration /Looping

1. Sequence Control Structure

- This is the default flow of a program.
- Statements execute one after another, from top to bottom, in the order they are written.
- Each step runs sequentially—no skipping, no repeating.

2. Selection (Decision-Making) Control Structure

- Here, the program chooses between different paths based on a condition.
- If a condition is true → one block runs & If false → another block runs.
- e.g. →
 - if
 - if-else
 - else if
 - switch

3. Iteration (Looping) Control Structure

- This is used when a block of code must be repeated multiple times.
- The same statements run again and again until a condition becomes false.
- e.g. →
 - for
 - while
 - do-while

Conditional Statements

- The conditional statements (also known as decision control structures) such as if, if else, switch, etc. are used for decision-making purposes in C++ programs.
- They are also known as Decision-Making Statements and are used to evaluate one or more conditions and make the decision whether to execute a set of statements or not.
- These decision-making statements in programming languages decide the direction of the flow of program execution.

❖ Need of Conditional Statements

- ✓ There come situations in real life when we need to make some decisions and based on these decisions, we decide what should we do next.
- ✓ Similar situations arise in programming also where we need to make some decisions and based on these decisions we will execute the next block of code.

Types of Conditional Statements in C++

- I. if statement
- II. if-else statement
- III. nested if-else statements
- IV. else-if ladder
- V. switch case statement
- VI. break statement

1. if in C++

- The if statement is the most simple decision-making statement.
- It is used to decide whether a certain statement or block of statements will be executed or not i.e if a certain condition is true then a block of statements is executed otherwise not.

Syntax :-

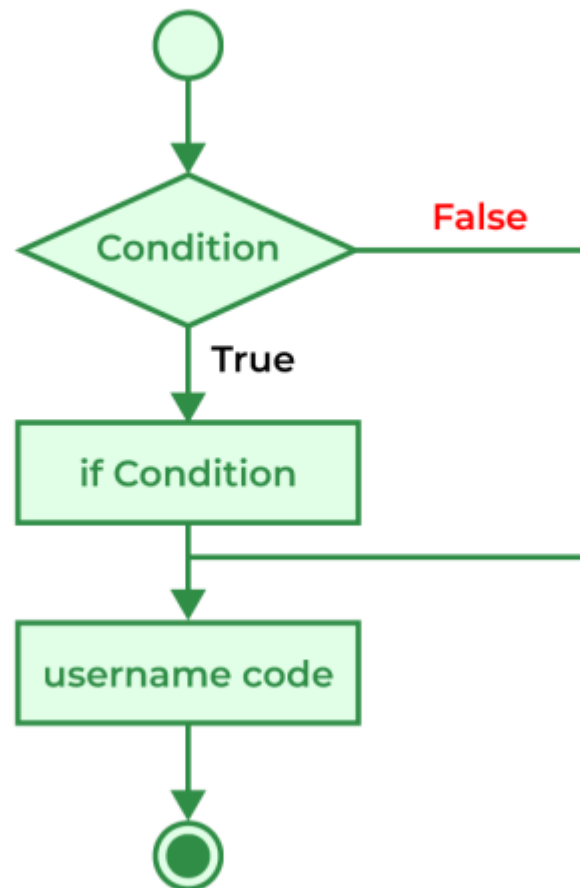
```
if(condition)
```

```
{
```

```
    Statements execute ; // if condition is true
```

```
}
```

Flowchart:



Question

- Q1.** Write a program in c++,in which Check person can apply for Driving licence or not. (Using only if statement)
- Q2.** Write a program in c++,in which Check person can eligible for vote or not. (Using only if statement)
- Q3.** Write a program in c++ take any number and check this number is Odd or Even. (Using only if statement)
- Q4.** Write a program in c++,in which Check Student pass or fail.(Using only if statement)
- Q5.** Write a program in c++ to verify whether an ATM withdrawal can be done based on available balance using only if statement.
- Q6.**Write a program to check the temperature condition of the day (hot or normal) using only if statement.
- Q7.** Write a program take two number and check greatest,smallest or equal.
- Q8.** Write a c++ program that takes the session year and the current year as input and tells which year of study the student is in.

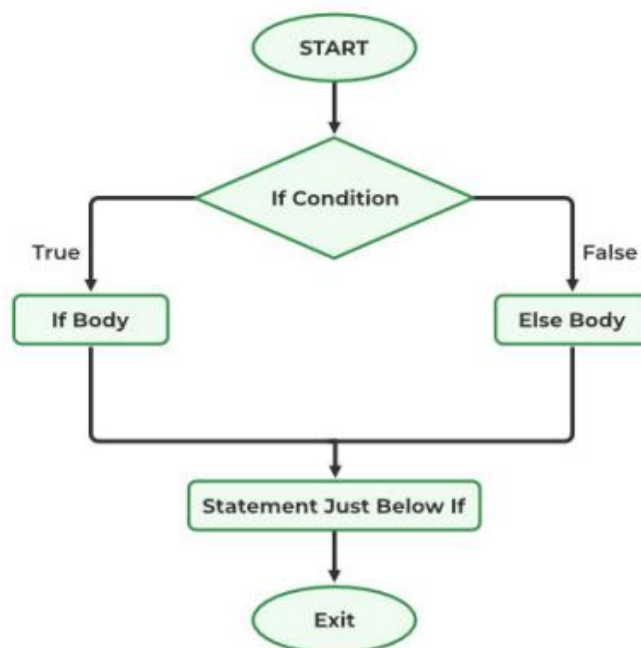
2. if-else in C++

- The if statement alone tells us that if a condition is true it will execute a block of statements and if the condition is false it won't. But what if we want to do something else when the condition is false?
- Here comes the C++ else statement. We can use the else statement with the if statement to execute a block of code when the condition is false.
- The if-else statement consists of two blocks, one for false expression and one for true expression.

Syntax :

```
if (condition)
{
    // Executes this block; // if condition is true
}
else
{
    // Executes this block; // if condition is false
}
```

Flowchart:



Question

Q9. Write a program in c++, in which compare two numbers and print which one is greater.

Q10. Write a program in c++, in which check whether a given number is even or odd using bitwise operators.

Q11. Write a program in c++, in which Create a login validation program.

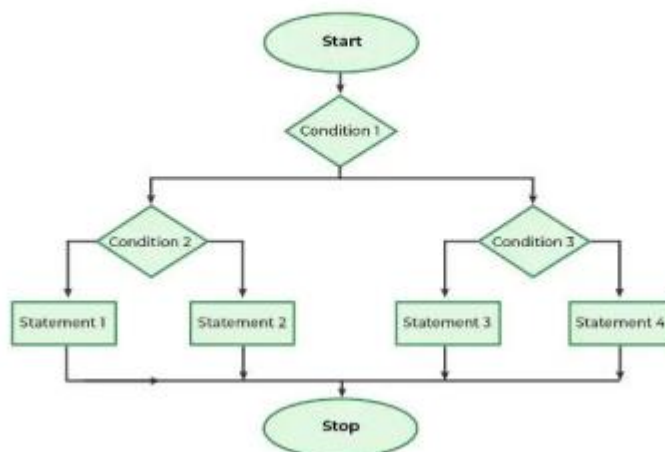
3. Nested if-else in C++

- A nested if in C++ allow us to nested if statements within if statements, i.e, we can place an if statement inside another if statement.

Syntax :

```
if (condition1)
{
    // Executes when condition1 is true
}
if (condition2)
{
    // Executes when condition2 is true
}
else
{
    // Executes when condition2 is false
}
```

Flowchart:



Question

Q12. Write a c++ program to input marks of a student and display the grade using nested if-else.

Q13. Write a c++ program to input three numbers and find the largest number using nested if-else.

Q14. Write a c++ program to check whether a year is a leap year or not using nested if-else.

Q15. Write a c++ program to input age and determine the ticket category (child,adult,senior) using nested if-else.

Q16. Write a c++ program to input a character and check whether it is a vowel, consonant,digit or special symbol using nexted if-else.

4. if-else-if Ladder in C++

- The if else if statements are used when the user has to decide among multiple options.
- The C++ if statements are executed from the top down. As soon as one of the conditions controlling the if is true, the statement associated with that if is executed, and the rest of the C++ else-if ladder is bypassed.
- If none of the conditions is true, then the final else statement will be executed. if-else-if ladder is similar to the switch statement.

Syntax :

```
if (condition)
```

```
    statement;
```

```
else if (condition)
```

```
    statement;
```

```
else if (condition)
```

```
    statement;
```

```
else if (condition)
```

```
    statement;
```

```
..
```

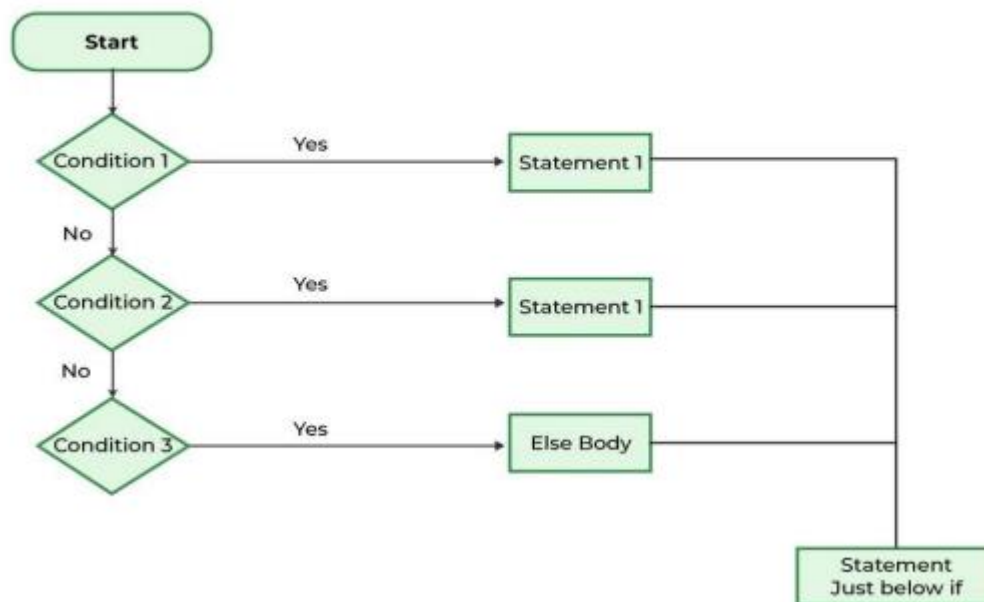
```
..
```

```
..
```

```
..else
```

```
    statement;
```

Flowchart:



Question

Q16. Write a c++ program to read marks of a student and print the 1st/2nd/3rd or grade using if-else-if ladder.

Q17. Write a c++ program to find the largest among three numbers using if-else-if ladder.

Q18. Write a c++ program to categorize electricity usage as low, medium, or high if-else-if ladder.

Q19. Write a c++ program to determine whether a person is a minor, adult, or senior citizen using if-else-if ladder.

Q20. Write a c++ program to check whether a number is positive, negative or zero using if-else-if ladder.

5. Break Statement

The break statement in c++ is used to immediately stop the execution of a loop or a switch statement and transfer control to the statement that comes after the loop or switch.

It is mainly used inside :- for loop, while loop, do-while loop, switch statement.

Syntax

Break;

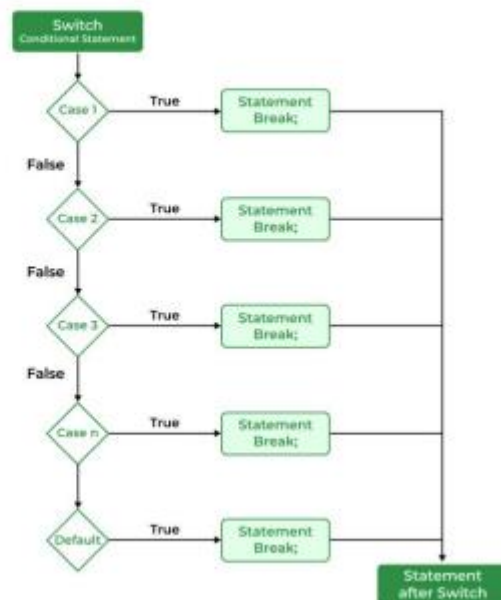
6. switch Statement in C++

The switch case statement is an alternative to the if else if ladder that can be used to execute the conditional code based on the value of the variable specified in the switch statement. The switch block consists of cases to be executed based on the value of the switch variable.

Syntax:

```
switch (expression)
{
case value1:
    statements;
case value2:
    statements;
.... ....
default:
    statements;
}
```

Flowchart:



Question

Q21. Write a c++ program to perform a basic calculator operation using switch statement.

- Q22.** Write a c++ program to display the name of the day based on day number using switch statement.
- Q23.** Write a c++ program to check whether a character is a vowel or consonant using switch statement.
- Q24.** Write a c++ program to print the month name based on month number using switch statement.
- Q25.** Write a c++ program to display a menu-driven program for area calculation of different shapes using switch statement.